

Let $f(x) \in F[x]$ be given with the irreducible factorization

$$f(x) = f_1(x) \cdots f_n(x).$$

Since every Automorphism fixing F maps root of f_i to the root of f_i , the same irreducible Polynomial, therefore, for the splitting field K of f ,

$$\text{Aut}(K/F) \hookrightarrow S_{f_1} \times \cdots \times S_{f_n}.$$

With such an observation, in this paper, we figure out concretely the Galois group of the polynomials.

Def. Let $x_1, \dots, x_n$ be indeterminates.

The elementary symmetric functions $S_1, S_2, \dots, S_n$ are defined by

$$S_1 = x_1 + x_2 + \dots + x_n$$

$$S_2 = x_1 x_2 + x_1 x_3 + \dots + x_1 x_n + x_2 x_3 + x_2 x_4 + \dots + x_{n-1} x_n$$

$\vdots$

$$S_n = x_1 x_2 \dots x_n$$

And, the general polynomial of degree n is the polynomial

$$f(x) = (x - x_1)(x - x_2) \dots (x - x_n).$$

Note that

$$f(x) = x^n - S_1 x^{n-1} + S_2 x^{n-2} - \dots + (-1)^n S_n$$

Thm. Let $x_1, \dots, x_n$ be indeterminates of a field F , and $S_1, \dots, S_n$ be symmetric functions. Then, $F(x_1, \dots, x_n)$ is Galois extension of $F(S_1, \dots, S_n)$ with

$$\text{Gal}(F(x_1, \dots, x_n) / F(S_1, \dots, S_n)) \cong S_n.$$

Proof. Since $F(x_1, \dots, x_n)$ is the splitting field of the polynomial

$$f(x) = (x - x_1) \dots (x - x_n) = x^n - S_1 x^{n-1} + \dots + (-1)^n S_n$$

over $F(S_1, \dots, S_n)$ and f is separable, so the first assertion is obtained.

And, since degree of f is n , so $[F(x_1, \dots, x_n) : F(S_1, \dots, S_n)] \leq n!$.

Meanwhile, for any permutation $\sigma \in S_n$, a induced map

$$\bar{\sigma} : F(x_1, \dots, x_n) \rightarrow F(x_1, \dots, x_n) : \frac{g(x_1, \dots, x_n)}{f(x_1, \dots, x_n)} \mapsto \frac{g(x_{\sigma(1)}, \dots, x_{\sigma(n)})}{f(x_{\sigma(1)}, \dots, x_{\sigma(n)})}$$

is an Automorphism which fixes $F(S_1, \dots, S_n)$, so

$$S_n \hookrightarrow \text{Aut}(F(x_1, \dots, x_n) / F(S_1, \dots, S_n))$$

Since $|S_n| = n!$, so proved. @

Discriminant.

Define the discriminant D of $x_1, \dots, x_n$ by the formula

$$D = \prod_{1 \leq i < j \leq n} (x_i - x_j)^2 \in \mathbb{Z}[x_1, \dots, x_n]$$

Define the discriminant of a polynomial to be the discriminant of the roots.

Using the discriminant D , denote that

$$\sqrt{D} := \prod_{1 \leq i < j \leq n} (x_i - x_j) \in \mathbb{Z}[x_1, \dots, x_n].$$

Note: the discriminant D is a symmetric function in $x_1, \dots, x_n$.

And, for a permutation $\sigma \in S_n (= \text{Sym}(x_1, \dots, x_n))$

$\sigma \in A_n$ if and only if σ preserves the sign of $\sqrt{D}$, that is, $\sigma(\sqrt{D}) = \sqrt{D}$.

Now, we obtain the following result:

Thm. Let F be a field which is not characteristic 2.

Then, for indeterminates $x_1, \dots, x_n$,

$$F(s_1, \dots, s_n) \leq_{\deg 2} F(s_1, \dots, s_n)(\sqrt{D}) = F(x_1, \dots, x_n) A_n \leq_{\deg n!/2} F(x_1, \dots, x_n)$$

Moreover, from the language of the polynomial, for

$\text{Gal}(K/F) \cong A_n$ if and only if $\sqrt{D}$ is fixed by all elements in $\text{Gal}(K/F)$.

Polynomials of degree 2.

Consider the polynomial $x^2 + ax + b$ with roots x_1 and x_2 .

The discriminant is

$$D = (x_1 - x_2)^2 = (x_1 + x_2)^2 - 4x_1x_2 = S_1^2 - 4S_2 = (-a)^2 - 4(b) = a^2 - 4b.$$

The polynomial is separable if and only if $x_1 \neq x_2$ if and only if $a^2 - 4b \neq 0$.

Since K , the splitting field of $x^2 + ax + b$ over $\mathbb{Q}$, $\text{Gal}(K/\mathbb{Q}) \hookrightarrow S_2$ and

$\text{Gal}(K/\mathbb{Q})$ is trivial if and only if $\sqrt{a^2 - 4b} \in \mathbb{Q}$.

$\text{Gal}(K/\mathbb{Q}) \cong C_2$ if and only if $\sqrt{a^2 - 4b} \notin \mathbb{Q}$.

Polynomial of degree 3, Cubic.

If the cubic polynomial is reducible, then it splits either into three linear factors or a linear factor $\times$ an irreducible quadratic.

In the first case, the Galois group is trivial, and in the second case, the Galois group is C_2 .

Assume that the polynomial f is irreducible.

Let K/F be the splitting field of the cubic.

Since $[K:F] \geq \deg f = 3$, so that there are only two possibilities,

$$\text{Gal}(K/F) \cong A_3 \text{ or } S_3.$$

Using the knowledge of the discriminant,

$$\text{Gal}(K/F) \cong A_3 \text{ if and only if } \sqrt{D} \in F.$$

